LESSON 7.3

Understand Absolute Value of Rational Numbers

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

WHY THIS MATTERS Claire is performing a science experiment with two liquids. The temperatures of the two liquids in Celsius are shown. She needs to use the beaker with the temperature that is closer to 0°C. Which beaker should Claire use?

REMEMBER

Understand Absolute Value of Rational Numbers

Absolute Value. Formula: |a| = Distance of a from 0 (always ≥ 0) A rational number is any number that can be written as a fraction: integers like -6, fractions like -3/4, and decimals like 2.5 are all rational numbers. The absolute value of a number is how far it is from zero on the number line. Distance is never negative.

1. Absolute value represents distance on a number line, never direction.
2. Distance is always positive or zero: |-7| = 7 and |7| =
3. Context: Magnitude of temperature, debt, elevation regardless of sign

Words: **Integer *(Número entero)*** **Positive *(Positivo)*** **Negative *(Negativo)*** **Absolute value *(Valor absoluto)*** **Opposite *(Opuesto)***

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

What is |−3/4|?

(A) 0
(B) 3/4
(C) −3/4
(D) 4/3

HINT Mark −3/4 on a number line between −1 and 0.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which number sits between 2 and 3 on a number line?

(A) 1.9
(B) 2.4
(C) −2.5
(D) 3.2

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the opposite of 3/4?

(A) 0
(B) −3/4
(C) 3/4
(D) 4/3

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Between which two integers does −1/2 sit on a number line?

(A) 0 and 1
(B) −1 and 0
(C) −1/2 is not on the number line
(D) −2 and −1

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The opposite of a number sits the same distance from 0 but on the…

(A) other side of 0
(B) same side of 0
(C) exact same point
(D) far end of the line

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

The Missing Sign Mistake

1. **Problem** Find the absolute value of -9.
2. **Student's thinking** The number inside the bars is negative, so I should keep the negative sign in my answer.
3. **Student's answer** |-9| = -9

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Can you walk backward -5 steps? Distance is always zero or positive, no matter which direction you moved.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

SENTENCE STARTER The integer ___ is located ___ units from zero, so its absolute value is ___. Even though ___ is negative, its distance from zero is still ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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LEARNING TARGET I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

WHY THIS MATTERS Claire is performing a science experiment with two liquids. The temperatures of the two liquids in Celsius are shown. She needs to use the beaker with the temperature that is closer to 0°C. Which beaker should Claire use?

1 **Circle the letter of the best answer. Show how you know.**

A submarine is at -150 meters (below sea level) and a helicopter is at +200 meters (above sea level). A rescue team needs to know how far each is from sea level to plan a mission. What is |−150|?

(A) 0
(B) 150
(C) −150
(D) 300

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A submarine is at -150 meters (below sea level) and a helicopter is at +200 meters (above sea level). A rescue team needs to know how far each is from sea level to plan a mission. Which craft is FARTHER from sea level?

(A) The helicopter, because 200 > 150
(B) The submarine, because −150 < 200
(C) They are the same distance
(D) You cannot compare them

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A submarine is at -150 meters (below sea level) and a helicopter is at +200 meters (above sea level). A rescue team needs to know how far each is from sea level to plan a mission. Which craft is at the LOWER position?

(A) The helicopter, because 200 > 150

(B) They are equal

(C) The submarine, because 150 < 200

(D) The submarine, because −150 < 200

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

On a frosty morning at the mountain cache, the thermometer reads -8 degrees. What is the absolute value of -8?

(A) -8

(B) 0

(C) 8

(D) 16

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Two treasure sites report their temperatures: -5 degrees F at the Ice Cave and -2 degrees F at the Frozen Dock. Which site is COLDER?

(A) -5 degrees F (Ice Cave)

(B) -2 degrees F (Frozen Dock)

(C) They are the same

(D) Neither is below zero

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Captain Vega compares |-12| (a deep cave) and |7| (a hilltop). Which absolute value is GREATER?

(A) |7| because 7 is positive

(B) They are equal

(C) |-12| = -12, so |7| is greater

(D) |-12| because 12 > 7

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help